

AWS Active Directory Domain Join & OU Management Lab

Objective:

Configure a Windows client to communicate with an Active Directory Domain Controller, join the client to the lab.local domain, rename the computer, move the computer object to the correct Organizational Unit (OU), and verify successful domain communication.

Architecture:

- **AWS EC2 WINDOWS SERVER (DOMAIN CONTROLLER)**
- **AWS EC2 WINDOWS CLIENT**
- **ACTIVE DIRECTORY DOMAIN SERVICES**
- **DNS**
- **ORGANIZATIONAL UNITS**

Domain: lab.local

Step 1 – Verify AWS Lab Environment

Verify that the required EC2 instances were running. PC-1 hosts the Active Directory Domain Controller, PC-2 is the client machine that will join the domain, and PC-3 is available for future use.

The screenshot displays the AWS Management Console's EC2 Instances page. The 'Instances' table shows the following data:

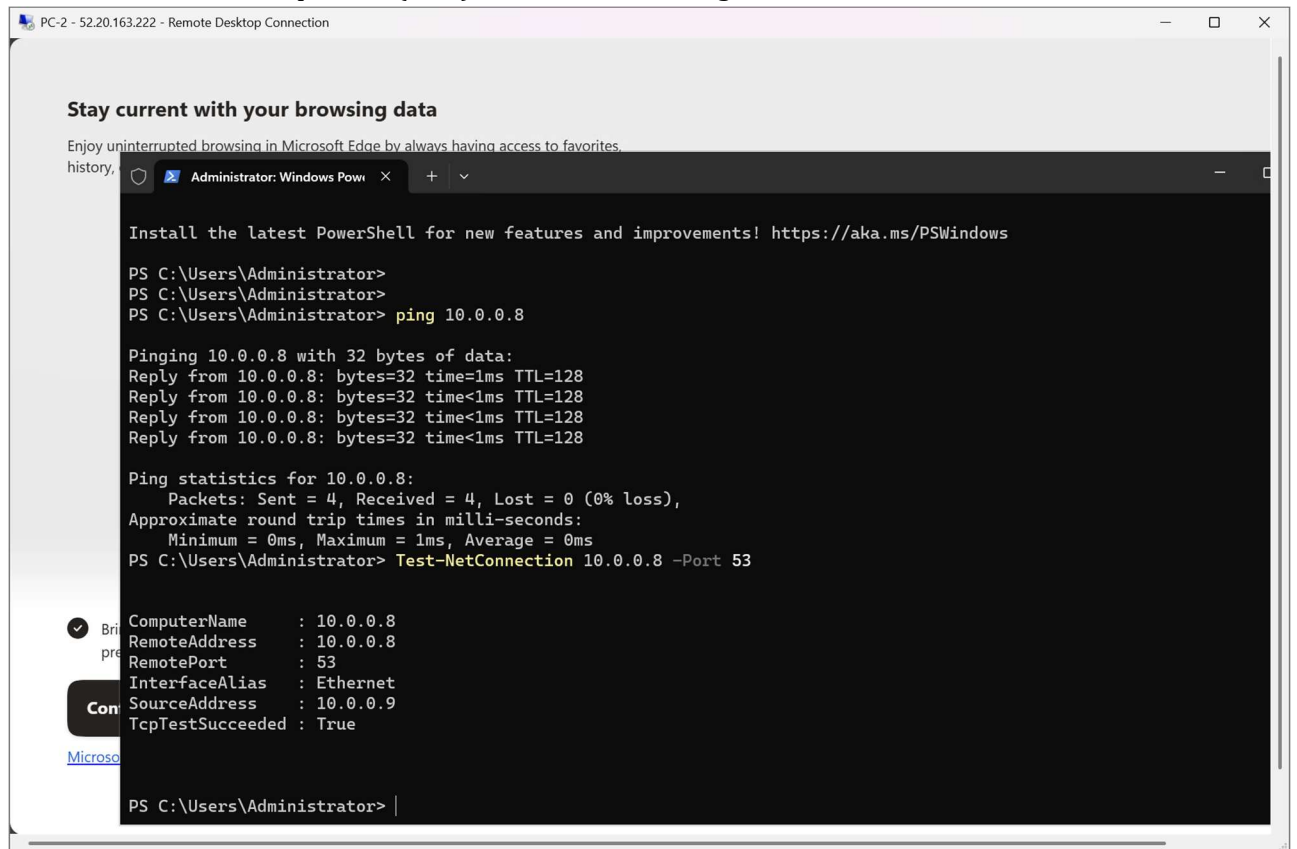
Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
devops-assign	i-0370bbaa7bf73c92f	Stopped	t3.micro	–	us-east-1c	ec2-52-194-153-128.co...	32.194.153.128	32.194.153.128	–
PC-1	i-0c354235b4496b468	Running	t3.micro	3/3 checks passed	us-east-1a	–	54.209.98.94	54.209.98.94	–
PC-3	i-0d96f3d23e4f9f5a9	Running	t3.micro	3/3 checks passed	us-east-1a	–	100.60.131.20	100.60.131.20	–
PC-2	i-06298fc1603a505ed	Running	t3.micro	3/3 checks passed	us-east-1a	–	52.20.163.222	52.20.163.222	–

The details for instance **i-0d96f3d23e4f9f5a9 (PC-3)** are shown below:

- Instance summary:**
 - Instance ID: i-0d96f3d23e4f9f5a9
 - IPv6 address: –
 - Hostname type: IP name: ip-10-0-0-10.ec2.internal
 - Answer private resource DNS name: –
 - Auto-assigned IP address: –
- Public IPv4 address:** 100.60.131.20 | [open address](#)
- Instance state:** Running
- Private IP DNS name (IPv4 only):** ip-10-0-0-10.ec2.internal
- Instance type:** t3.micro
- VPC ID:** vpc-0ebc4e9ef1d6213 (VPC-PC)
- Private IPv4 addresses:** 100.0.0.10
- Public DNS:** –
- Elastic IP addresses:** 100.60.131.20 [Public IP]
- AWS Compute Optimizer finding:** Opt-in to AWS Compute Optimizer for recommendations. | [Learn more](#)

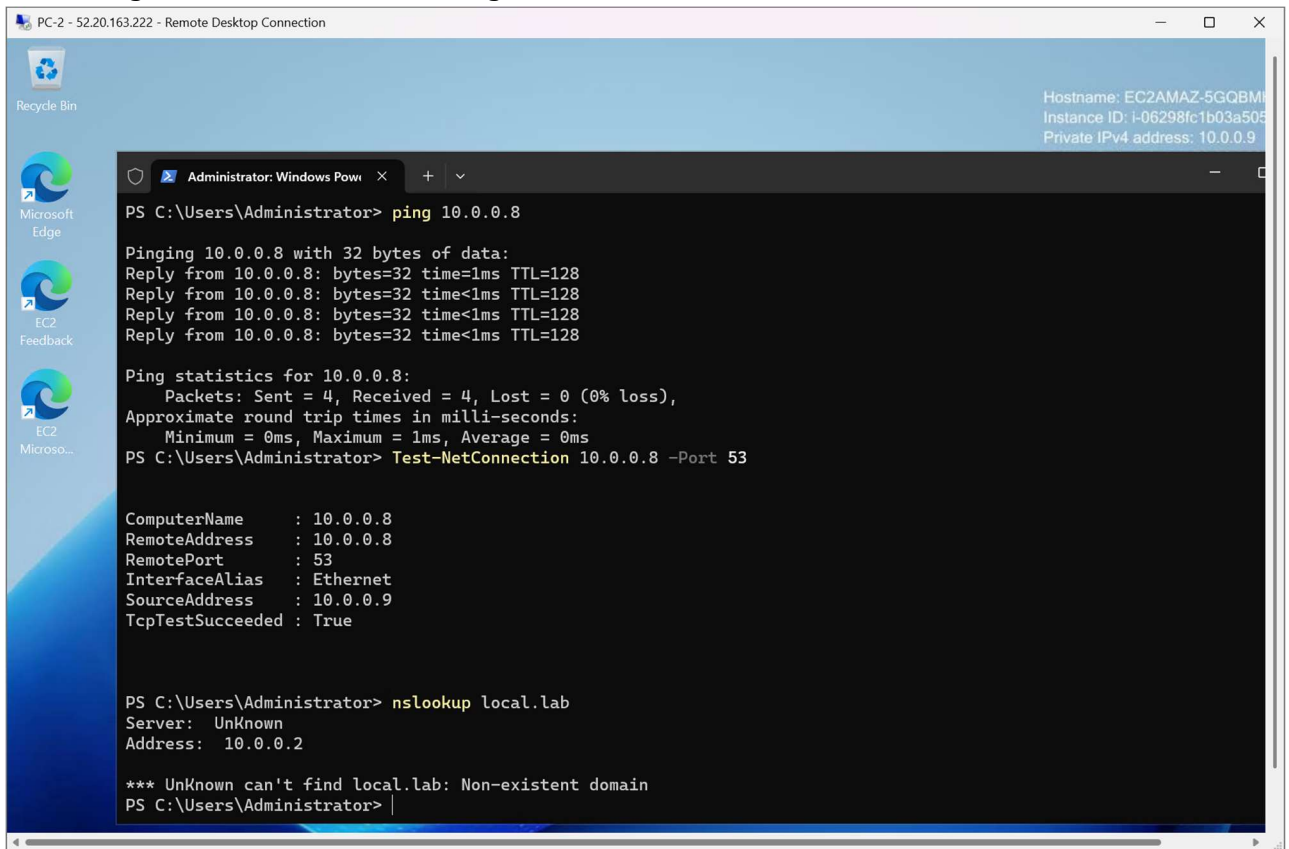
Step 2 – Test Basic Network Connectivity

From PC-2, verified Layer 3 connectivity to the Domain Controller (10.0.0.8) using ping. Then verified that TCP port 53 (DNS) was reachable using Test-NetConnection.



Step 3 – Identify the DNS Resolution Problem

Attempt to resolve the domain using nslookup. The query failed with 'Non-existent domain', confirming that the client was not using the correct DNS server.



```
PC-2 - 52.20.163.222 - Remote Desktop Connection

Recycle Bin
Microsoft Edge
EC2 Feedback
EC2 Microsoft...

Administrator: Windows PowerShell
PS C:\Users\Administrator> ping 10.0.0.8

Pinging 10.0.0.8 with 32 bytes of data:
Reply from 10.0.0.8: bytes=32 time=1ms TTL=128
Reply from 10.0.0.8: bytes=32 time<1ms TTL=128
Reply from 10.0.0.8: bytes=32 time<1ms TTL=128
Reply from 10.0.0.8: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
PS C:\Users\Administrator> Test-NetConnection 10.0.0.8 -Port 53

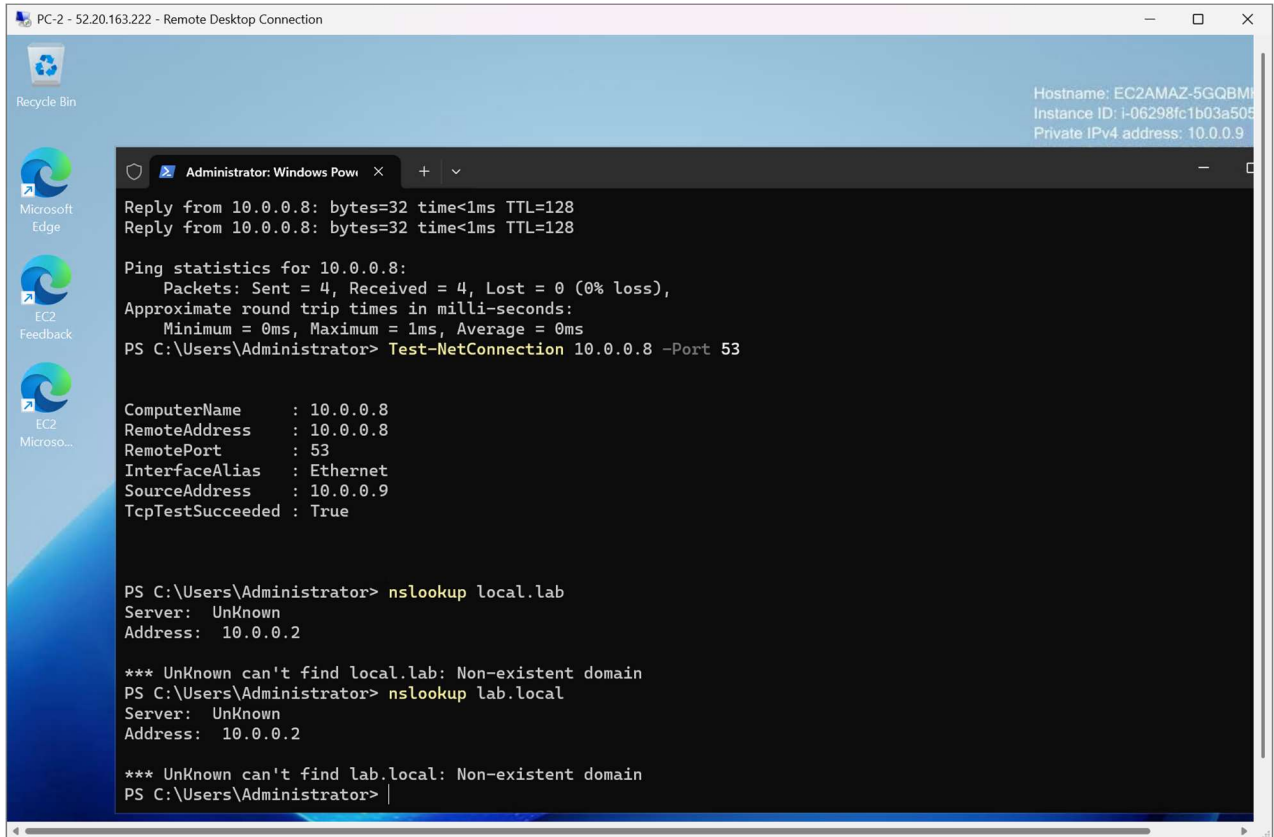
ComputerName      : 10.0.0.8
RemoteAddress     : 10.0.0.8
RemotePort        : 53
InterfaceAlias    : Ethernet
SourceAddress     : 10.0.0.9
TcpTestSucceeded  : True

PS C:\Users\Administrator> nslookup local.lab
Server: UnKnown
Address: 10.0.0.2

*** UnKnown can't find local.lab: Non-existent domain
PS C:\Users\Administrator>
```

Step 4 – Verify Client Network Configuration

Execute `ipconfig /all` to inspect the client's IP configuration. Confirmed that the DNS server was incorrectly configured as 10.0.0.2 instead of the Domain Controller.



```
PC-2 - 52.20.163.222 - Remote Desktop Connection

Recycle Bin
Microsoft Edge
EC2 Feedback
EC2 Microsoft...

Administrator: Windows PowerShell

Reply from 10.0.0.8: bytes=32 time<1ms TTL=128
Reply from 10.0.0.8: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
PS C:\Users\Administrator> Test-NetConnection 10.0.0.8 -Port 53

ComputerName      : 10.0.0.8
RemoteAddress     : 10.0.0.8
RemotePort        : 53
InterfaceAlias    : Ethernet
SourceAddress     : 10.0.0.9
TcpTestSucceeded  : True

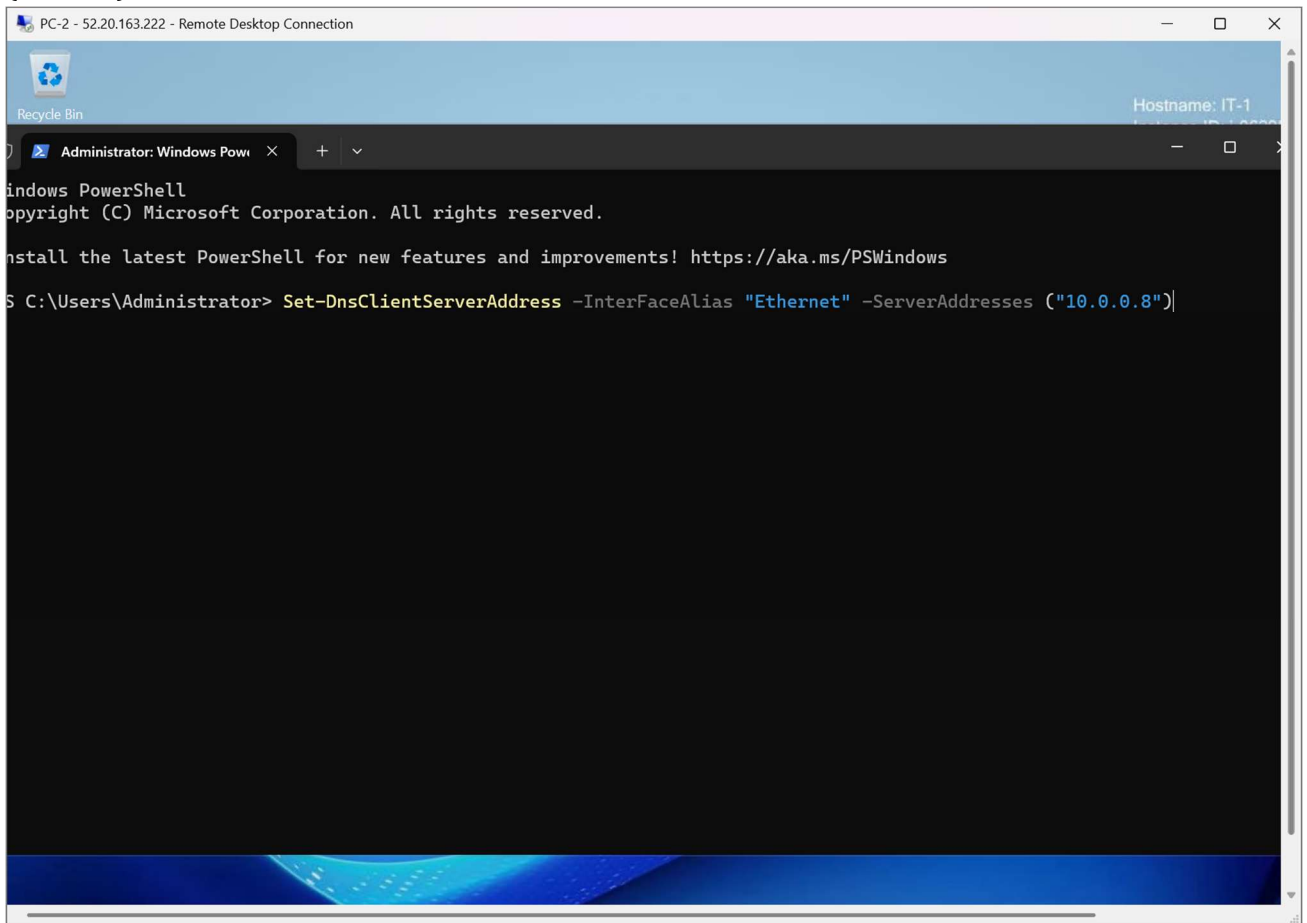
PS C:\Users\Administrator> nslookup local.lab
Server: UnKnown
Address: 10.0.0.2

*** UnKnown can't find local.lab: Non-existent domain
PS C:\Users\Administrator> nslookup lab.local
Server: UnKnown
Address: 10.0.0.2

*** UnKnown can't find lab.local: Non-existent domain
PS C:\Users\Administrator>
```

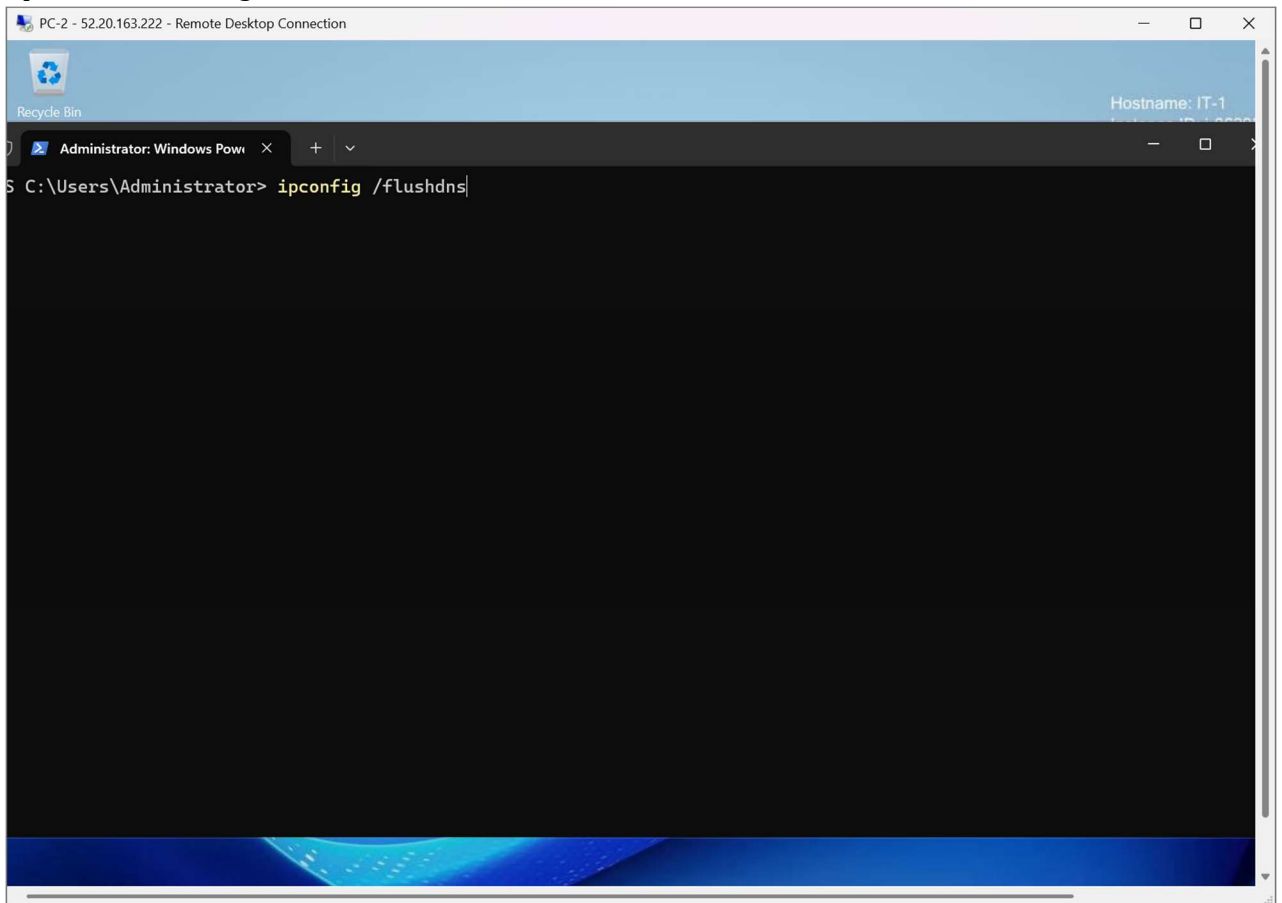
Step 5 – Configure the Correct DNS Server

Change the Preferred DNS Server on the client to the Domain Controller IP address (10.0.0.8).



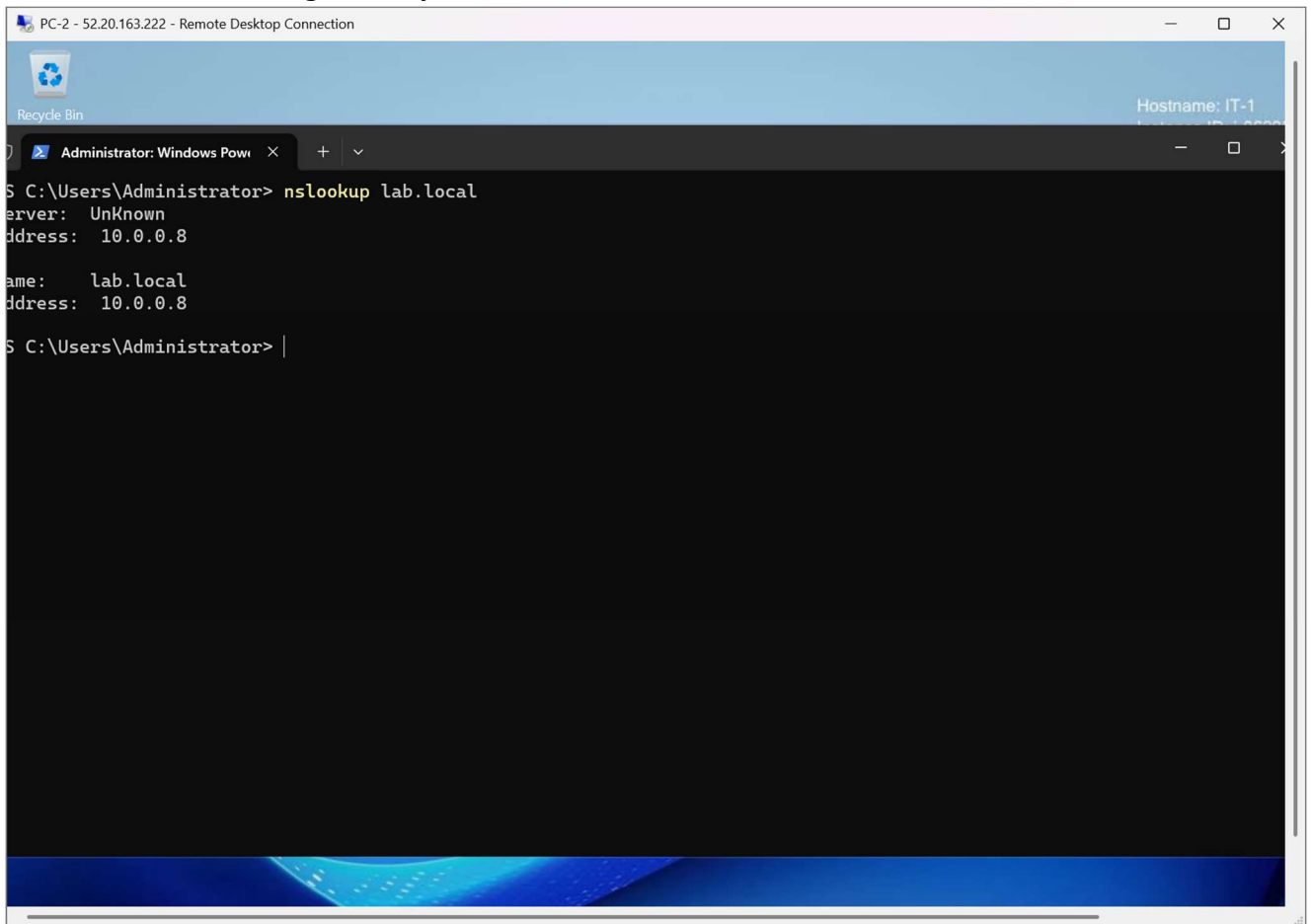
Step 6 – Clear Cached DNS Records

Execute `ipconfig /flushdns` to remove cached DNS entries so that future queries use the updated DNS configuration.



Step 7 – Verify DNS Resolution

Run `nslookup lab.local` again. The query now resolved successfully to 10.0.0.8, confirming that DNS was functioning correctly.



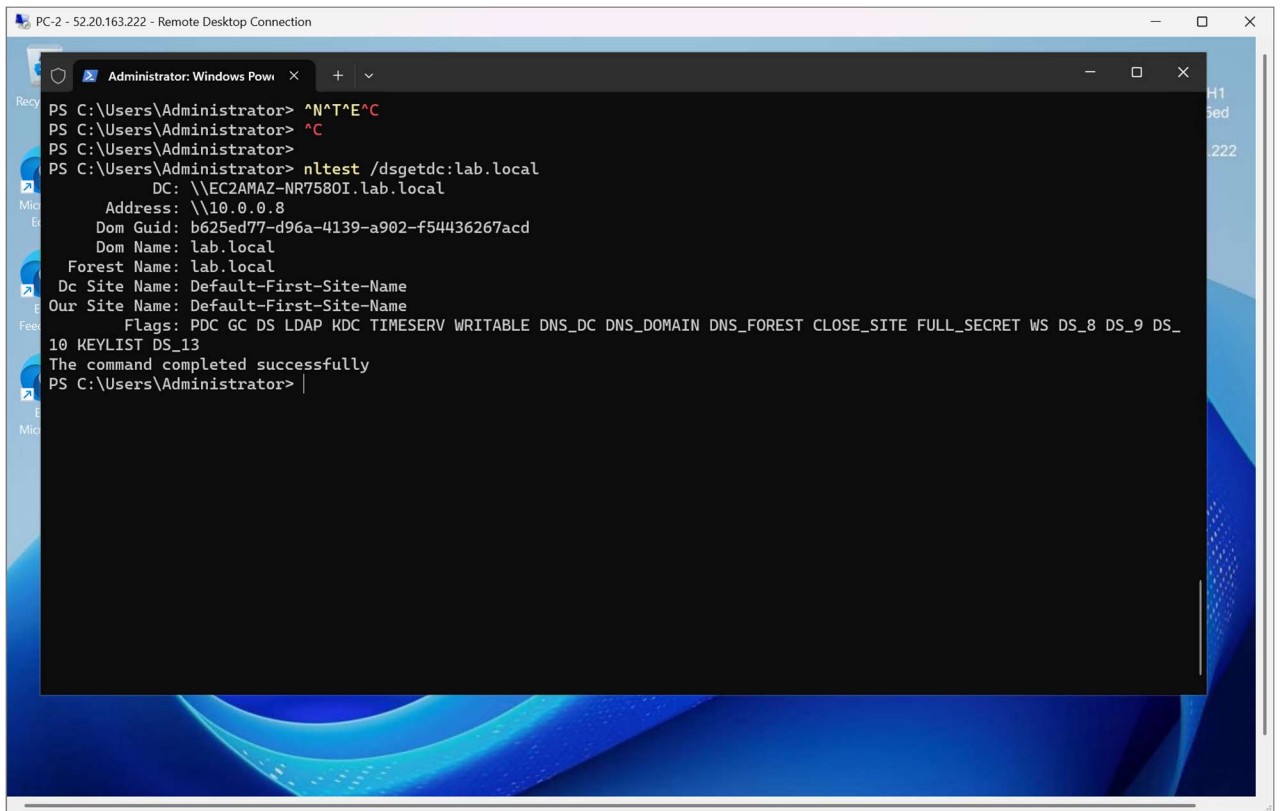
```
PC-2 - 52.20.163.222 - Remote Desktop Connection
Recycle Bin
Administrator: Windows PowerShell
C:\Users\Administrator> nslookup lab.local
Server: Unknown
Address: 10.0.0.8

Name: lab.local
Address: 10.0.0.8

C:\Users\Administrator> |
```

Step 8 – Verify Domain Controller Discovery

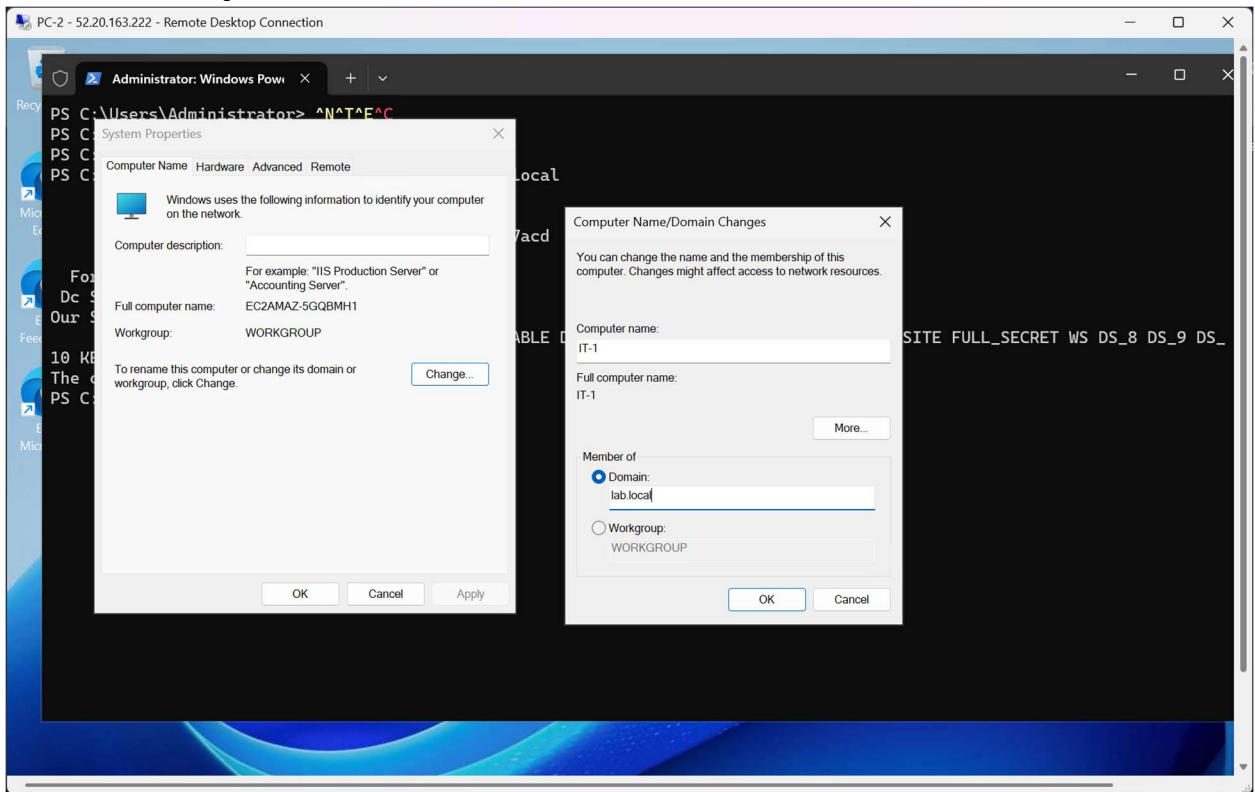
Execute `nltest /dsgetdc:lab.local` to verify that the client could locate the Active Directory Domain Controller.



```
PC-2 - 52.20.163.222 - Remote Desktop Connection
Administrator: Windows Pow
PS C:\Users\Administrator> ^N^T^E^C
PS C:\Users\Administrator> ^C
PS C:\Users\Administrator>
PS C:\Users\Administrator> nltest /dsgetdc:lab.local
        DC: \\EC2AMAZ-NR7580I.lab.local
        Address: \\10.0.0.8
        Dom Guid: b625ed77-d96a-4139-a902-f54436267acd
        Dom Name: lab.local
        Forest Name: lab.local
        Dc Site Name: Default-First-Site-Name
        Our Site Name: Default-First-Site-Name
        Flags: PDC GC DS LDAP KDC TIMESERV WRITABLE DNS_DC DNS_DOMAIN DNS_FOREST CLOSE_SITE FULL_SECRET WS DS_8 DS_9 DS_
10 KEYLIST DS_13
The command completed successfully
PS C:\Users\Administrator> |
```

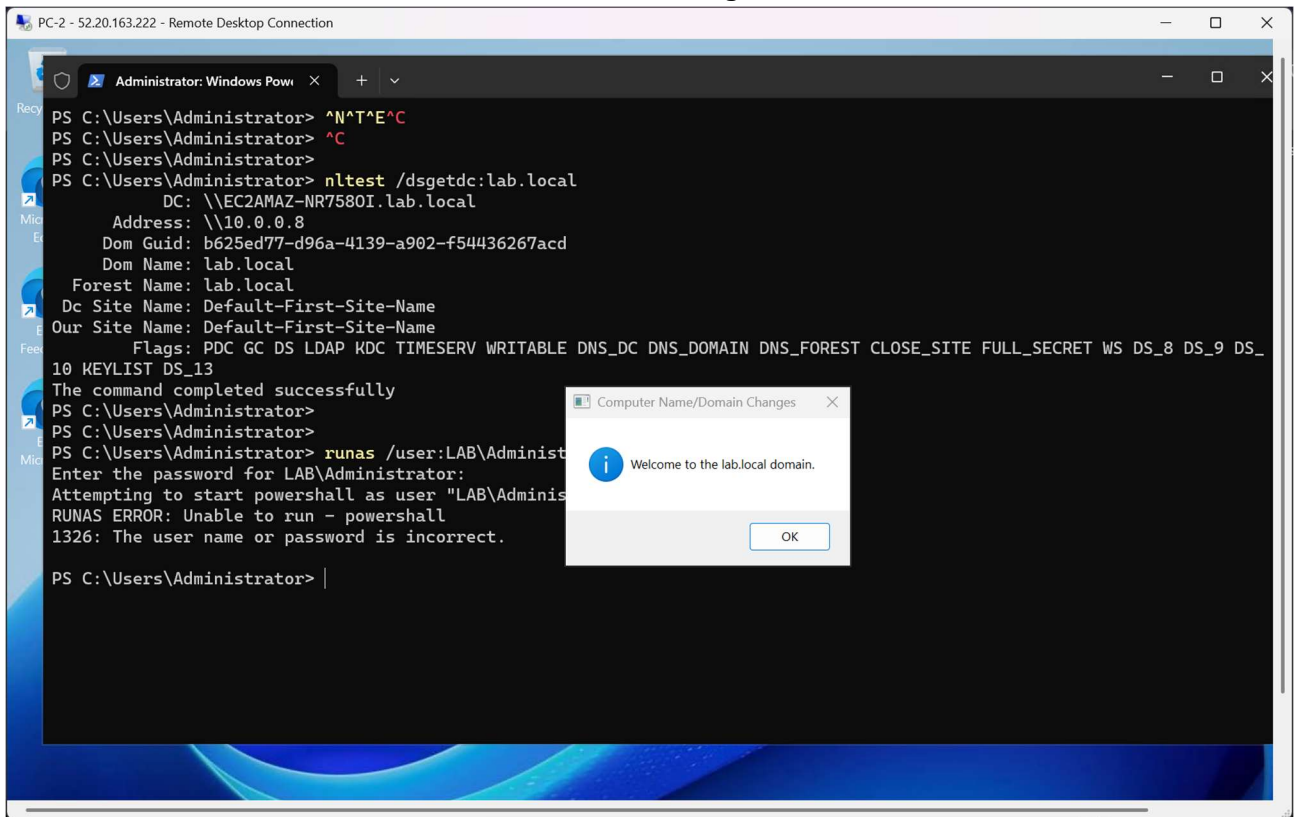

Step 9 – Join the Client to the Domain

Open System Properties, changed the computer membership from Workgroup to the lab.local domain, provided domain administrator credentials.



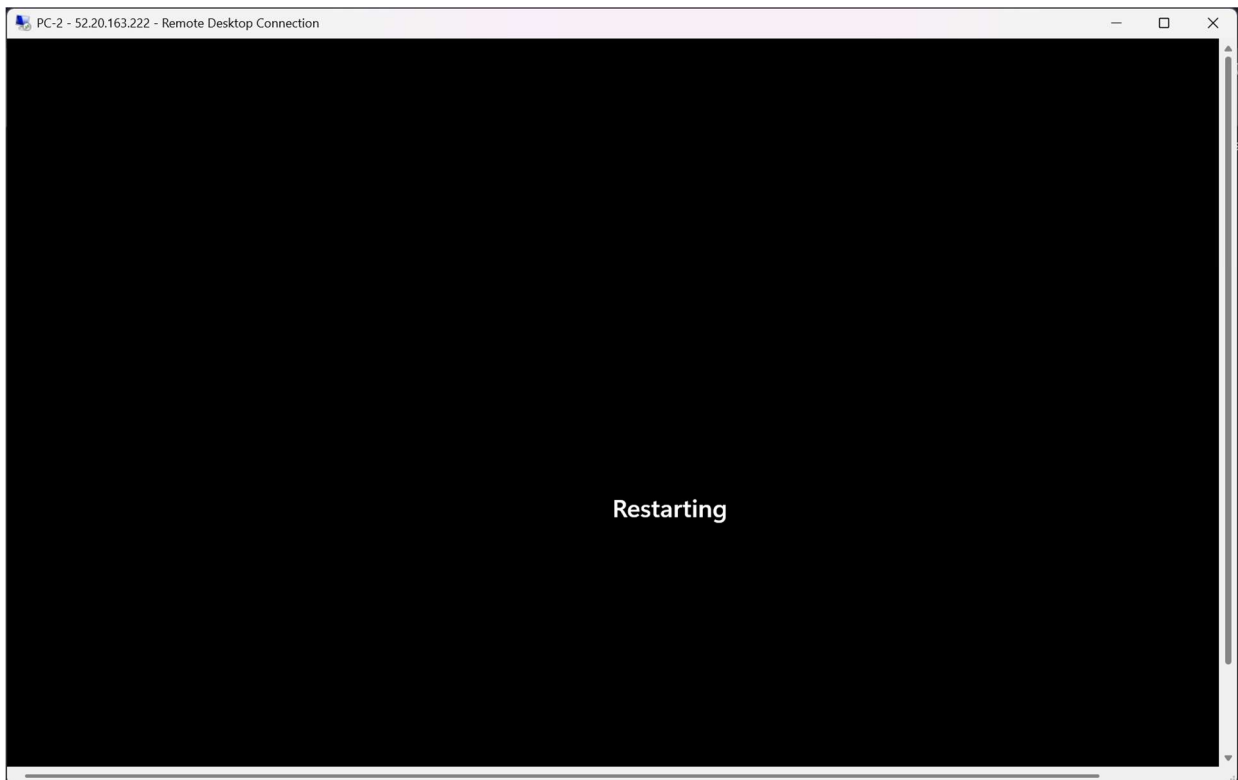
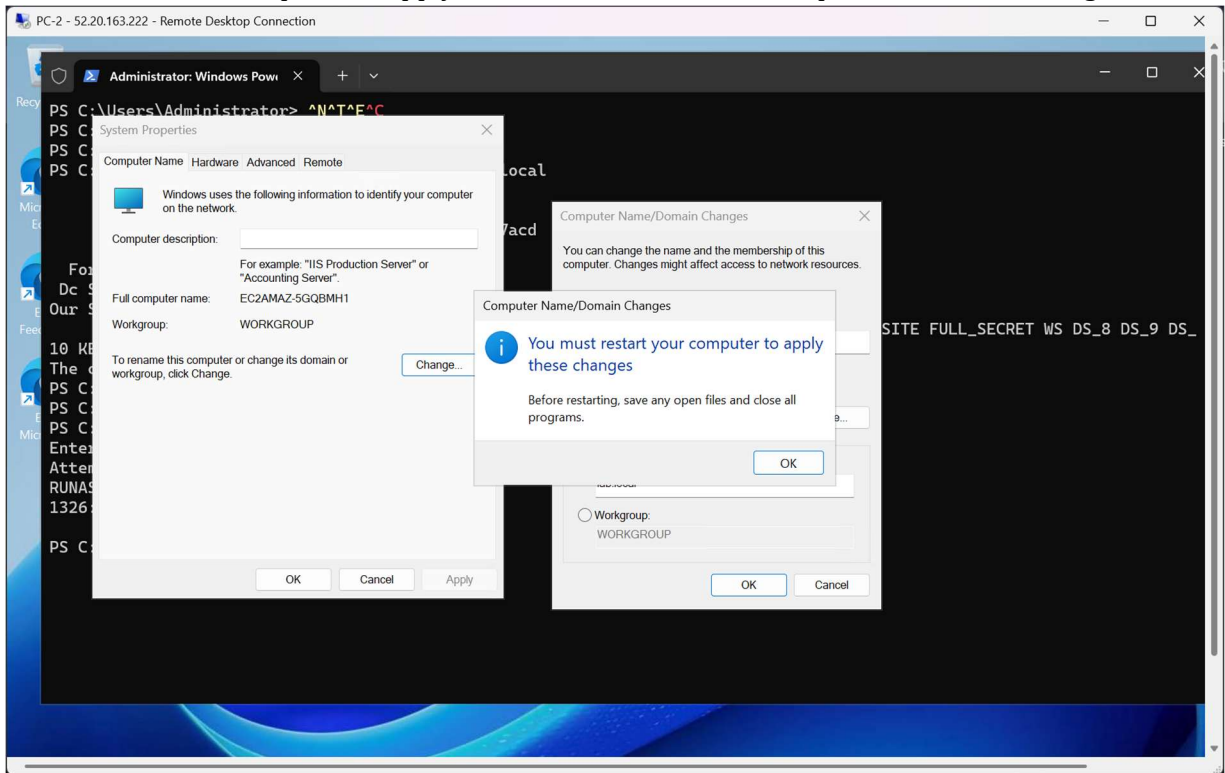
Step 10 – Rename the Client Computer

Change the default computer name to IT-1 while keeping it joined to the lab.local domain, , and received the 'Welcome to the lab.local domain' message



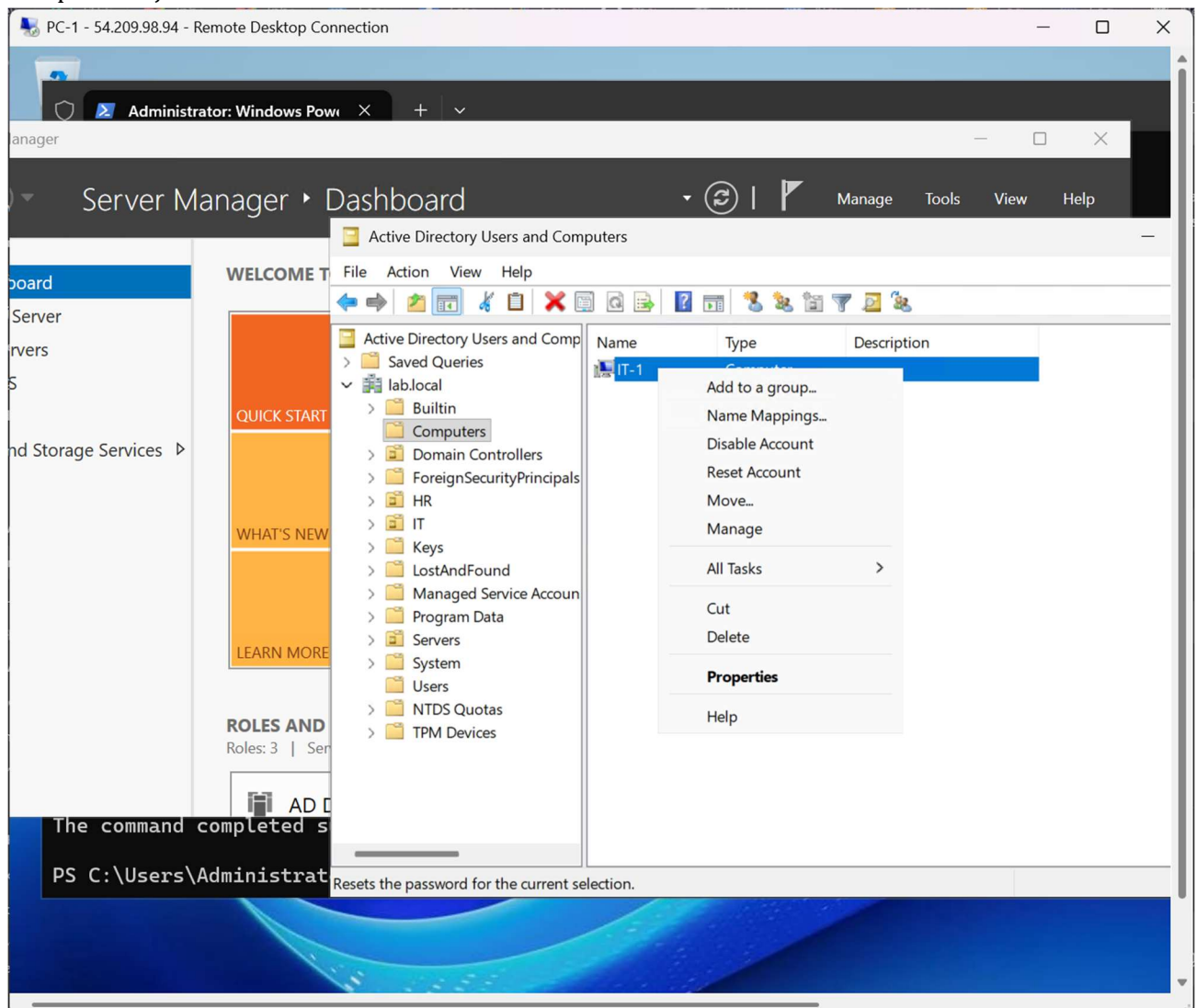
Step 11 – Restart the Client

Restart the client computer to apply both the domain membership and hostname changes.



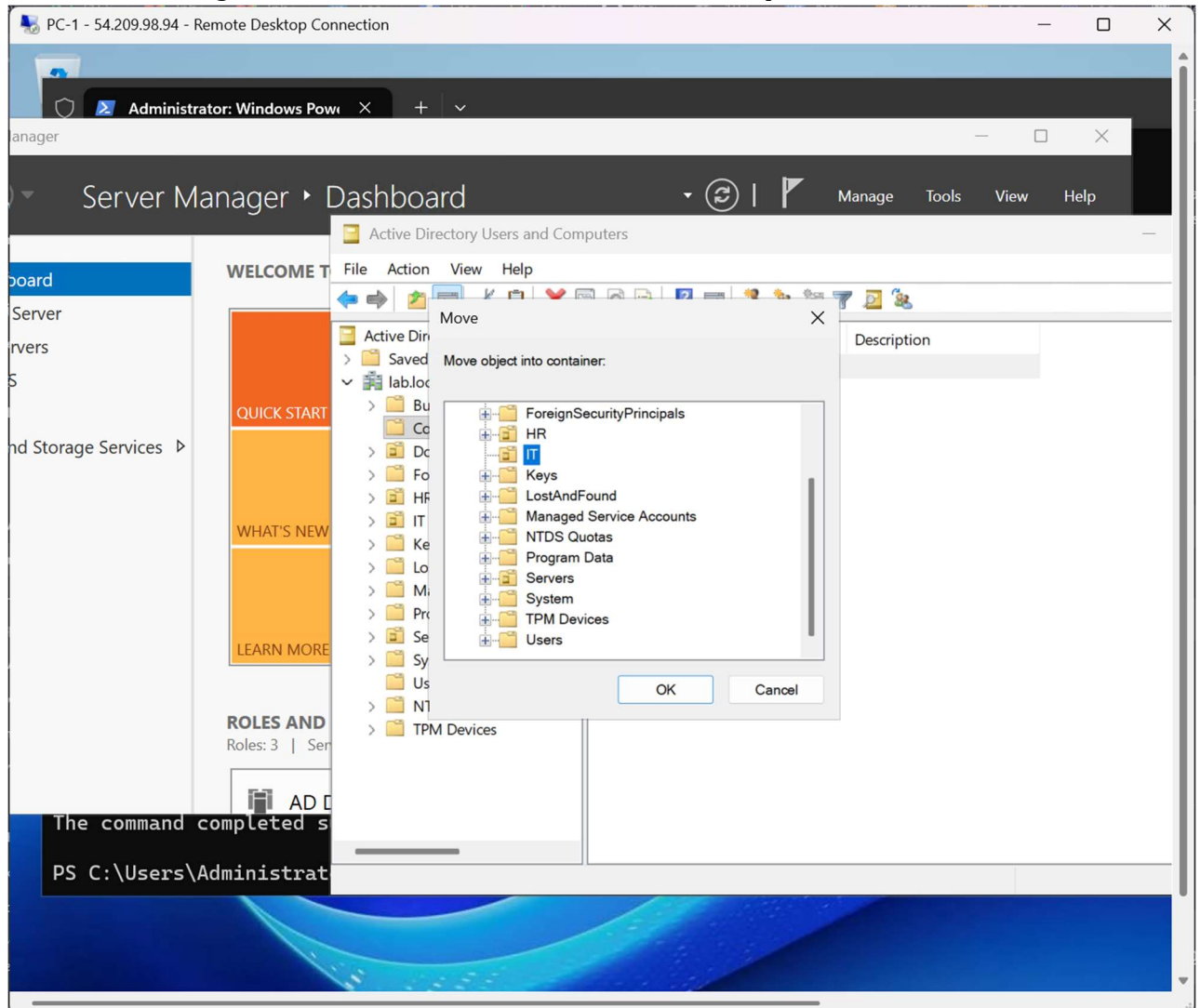
Step 12 – Move the Computer Object in Domain Controller

On the Domain Controller, open Active Directory Users and Computers, right-clicked the computer object and selected Move.



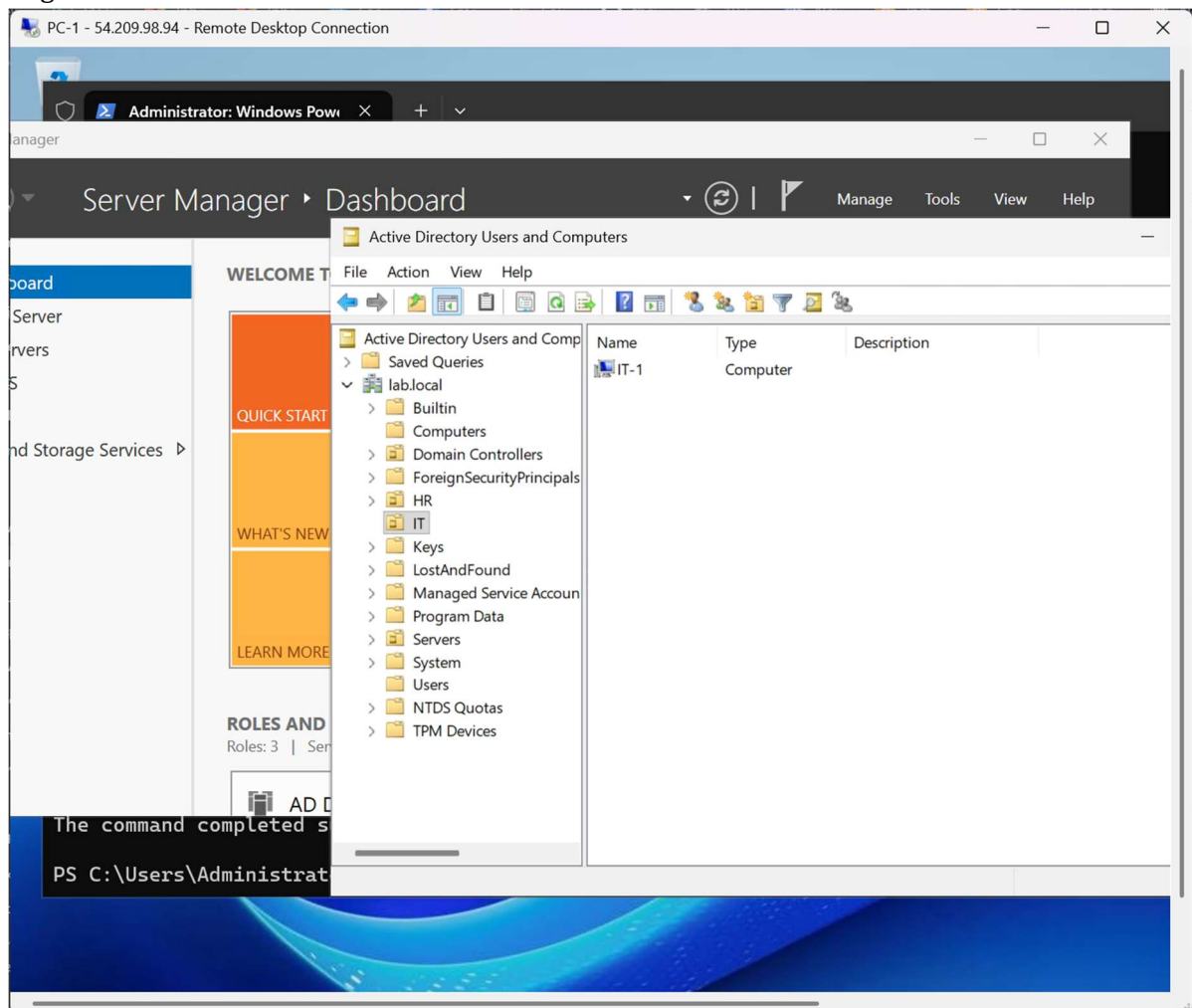
Step 13 – Select the Target OU

Selected the IT Organizational Unit as the destination for the computer account.



Step 14 – Verify OU Placement

Confirmed that the renamed computer (IT-1) appeared successfully inside the IT Organizational Unit.



Commands Used

- **PING 10.0.0.8**
- **TEST-NETCONNECTION 10.0.0.8 -PORT 53**
- **IPCONFIG /ALL**
- **SET-DNSCLIENTSERVERADDRESS -INTERFACEALIAS ETHERNET -SERVERADDRESSES 10.0.0.8**
- **IPCONFIG /FLUSHDNS**
- **NSLOOKUP LAB.LOCAL**
- **NLTEST /DSGETDC:LAB.LOCAL**

Conclusion

The client machine was successfully configured to use the Domain Controller as its DNS server, joined to the lab.local Active Directory domain, renamed to IT-1, moved into the IT Organizational Unit, and verified through DNS lookup and Active Directory management tools.